higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(Second Paper)
NQF LEVEL 2

7 NOVEMBER 2019

This marking guideline consists of 8 pages.

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QUESTION 1

1.1 1.1.1 The middle value of numerically arranged data. (1)

1.1.2 The value that occurs most often in a data set. (1)

1.2 1.2.1 27 (1)

1.2.2 $Mean = \frac{1591}{27}$ ✓

$= 58,926$ ✓ (2)

1.2.3

Stem	Leaf
3	0 0 6 9 ✓
4	1 3 4 8 ✓
5	1 2 4 4 6 6 6 6 ✓
6	1 3 7 ✓
7	0 5 6 7 ✓
8	3 4✓
9	0 7✓

(7)

1.2.4 Median = 56 (2)

1.2.5 Mode = 56 (1)

1.3 1.3.1 Min. = 7✓
Max. = 74✓ (2)

1.3.2 Arrange data in ascending order:
7 10 15 23 | 26 27 31 34 | 36 38 45 52 | 53 60 64 74

$Q_1(Position) = \frac{1}{4}(n+1)$

$= \frac{1}{4}(16+1)$

$= \frac{17}{4}$

$= 4,25$ ✓

$\therefore Q_1 = 23 + \frac{1}{4}(26 - 23)$

$= 23 + 0,75$

$= 23,75$✓

ALTERNATIVE: Using counting method:

$Q_1 = \frac{23+26}{2}$ ✓

$= 24,5$ ✓ (2)

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1.3.3 $Q_3(Position) = \frac{3}{4}(n+1)$

$= \frac{3}{4}(16+1)$

$= \frac{51}{4}$

$= 12,75$ ✓

$\therefore Q_3 = 52 + \frac{3}{4}(53-52)$

$= 52 + 0,75$

$= 52,75$ ✓

ALTERNATIVE: Using counting method:

$Q_1 = \frac{52+53}{2}$ ✓

$= 52,5$ ✓ (2)

1.3.4 $IQR = 52,75 - 23,75$ ✓

$= 29$ ✓

ALTERNATIVE: Using alternative answers in 1.3.3 and 1.3.4

$IQR = 52,5 - 24,5$ ✓

$= 28$ ✓

(2)

1.3.5 $i = \frac{P}{100} \times n$ ✓

$= \frac{60}{100} \times 16$

$= 9,6$ ✓

$\therefore 10\,th\ \ percentile = 38$ ✓

60th percentile =38

ALTERNATIVE:

$P_j(Position) = \frac{j}{100}(n+1)$

$P_{60}(Position) = \frac{60}{100}(16+1) = 10,2$ ✓ ✓

$P_{60} = 38 + 0,2(45-38) = 39,4$ ✓

Accept $P_{60} = 38$ (3)

1.3.6 Range = 74 – 7 ✓

= 67 ✓ (2)

1.4 1.4.1

MARKS	TALLY	FREQUENCY
$0 \le x < 20$	////	5

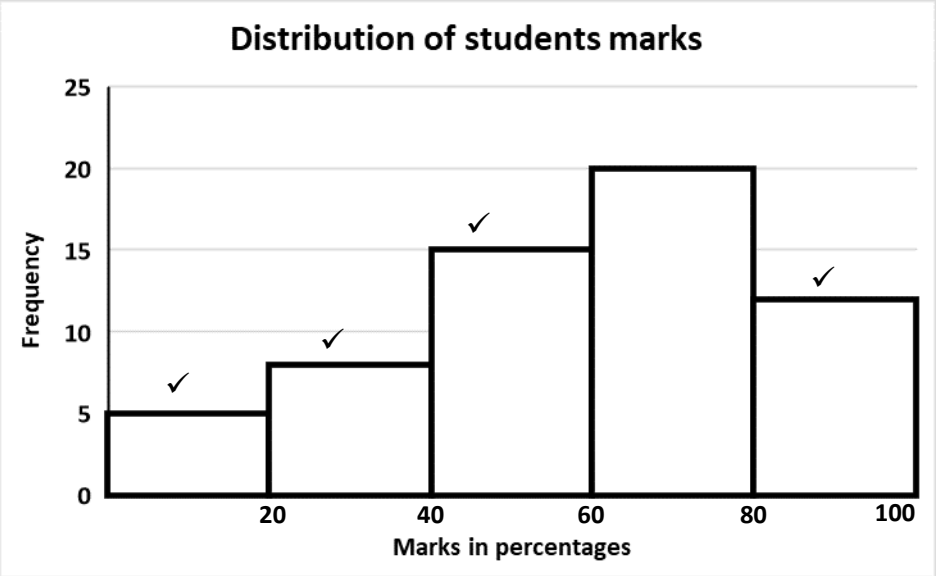
$20 \le x < 40$	𝍸 ///	8
$40 \le x < 60$	𝍸 𝍸 𝍸	15
$60 \le x < 80$	𝍸 𝍸 𝍸 𝍸	20
$80 \le x < 100$	𝍸 𝍸 //	12

(5)

1.4.2 $60 \le x < 80$ (1)

1.4.3 $60 \le x < 80$ (1)

1.4.4



Subtract 1 mark if a bar graph, not a histogram (5)
[40]

QUESTION 2

2.1 2.1.1 $E(\frac{-3+5}{2};\frac{2-2}{2})$ ✓
$E(1;0)$ ✓ (2)

2.1.2 $D(1=\frac{-2+x}{2};\ 0=\frac{-4+y}{2})$ ✓
$D(2=-2+x;\ 0=-4+y)$ ✓
$D(x=4;y=4)$
$D(4;4)$ ✓✓ (4)

2.1.3 $BC=\sqrt{(5+2)^2+(-2+4)^2}$ ✓
$=\sqrt{49+4}$ ✓
$=\sqrt{53}$ ✓ (3)

2.1.4
$$m_{AB} = \frac{2+4}{-3+2} \quad \checkmark$$
$$= \frac{6}{-1}$$
$$= -6 \checkmark$$
$$m_{DC} = \frac{4+2}{4-5} \checkmark$$
$$= \frac{6}{-1}$$
$$= -6 \quad \checkmark$$
(4)

2.2 2.2.1
$$P'(6;6) \checkmark\checkmark$$
$$Q'(1;3) \checkmark\checkmark$$
$$R'(9;1) \checkmark\checkmark$$
(6)

2.2.2 $P'(-3;4)$ (2)

2.3
$$V = \pi r^2 h$$
$$769,690 = \pi(7)^2 h \checkmark$$
$$h = \frac{769,690}{49\pi} \checkmark$$
$$h = 5cm \quad \checkmark$$
(3)

2.4 2.4.1
$$C = 2\pi r$$
$$= 2\pi(23cm) \checkmark \quad \checkmark$$
$$= 144,513cm \checkmark \quad \text{for correct radius}$$
(3)

2.4.2
$$A = \pi r^2$$
$$= \pi(23) \quad \checkmark \quad \checkmark$$
$$= 1661,903cm^2 \quad \checkmark$$

ALTERNATIVE
$$Area = \frac{\pi \times d^2}{4}$$
$$= \frac{\pi \times 46^2}{4} \quad \checkmark$$
$$= 1661,903 \ cm^2 \checkmark$$

(3)
[30]

QUESTION 3

3.1 3.1.1 $r^2 = (5)^2 + (12)^2$ ✓

$= 25 + 144$

$= 169$

$r = 13$ ✓ (2)

3.1.2 $= \frac{12}{13} - \frac{5}{13}$ ✓✓

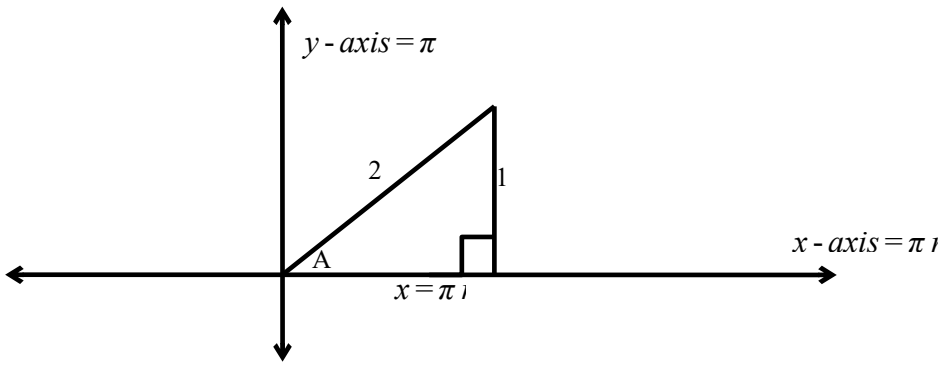
$= \frac{7}{13}$ ✓ (3)

3.1.3 $= \frac{12}{5} - \frac{7}{5}$ ✓

$= \frac{5}{5}$

$= 1$ ✓ (2)

3.2 3.2.1



$\sin A = \frac{1}{2}$

$r^2 = x^2 + y^2$

$(2)^2 = x^2 + (1)^2$ ✓

$4 = x^2 + 1$

$x^2 = 4 - 1$

$x^2 = 3$

$x = \sqrt{3}$

$\text{Tan A} = \frac{1}{\sqrt{3}}$ ✓ (3)

3.2.2 $= 1 - (\frac{1}{2})^2$ ✓

$= 1 - \frac{1}{4}$

$= \frac{3}{4}$ ✓ (2)

3.3 3.3.1 $\cos(55° + 35°)$ ✓

$\cos 90°$

$= 0$ ✓ (2)

3.3.2 $\cos 55° + \cos 35°$

$0{,}573576436 + 0{,}819152044$ ✓

$1{,}393$ ✓ (2)

3.4 3.4.1 $\tan A = \frac{BC}{AB} = \frac{2}{4{,}5}$ ✓

$\therefore \hat{A} = \tan^{-1} \frac{2}{4{,}5}$ ✓

$= 23{,}962°$ ✓ (3)

3.4.2 Using Pythagoras theorem:

$AC^2 = AB^2 + BC^2$

$AC^2 = 4{,}5^2 + 2^2$ ✓

$AC = \sqrt{20{,}25 + 4}$ ✓

$= 4{,}924 \ m$ ✓

ALTERNATIVE:

$\sin 24° = \frac{2}{AC}$ ✓

$AC = \frac{2}{\sin 23{,}962°}$ ✓

$= 4{,}925 \ m$ ✓

ALTERNATIVE:

$\cos 24° = \frac{4{,}5}{AC}$ ✓

$AC = \frac{4{,}5}{\sin 23{,}962°}$ ✓

$= 4{,}924 \ m$ ✓ (3)

3.5 3.5.1 $\hat{P} = 60° (\angle s \text{ of a } \Delta)$ (2)

3.5.2

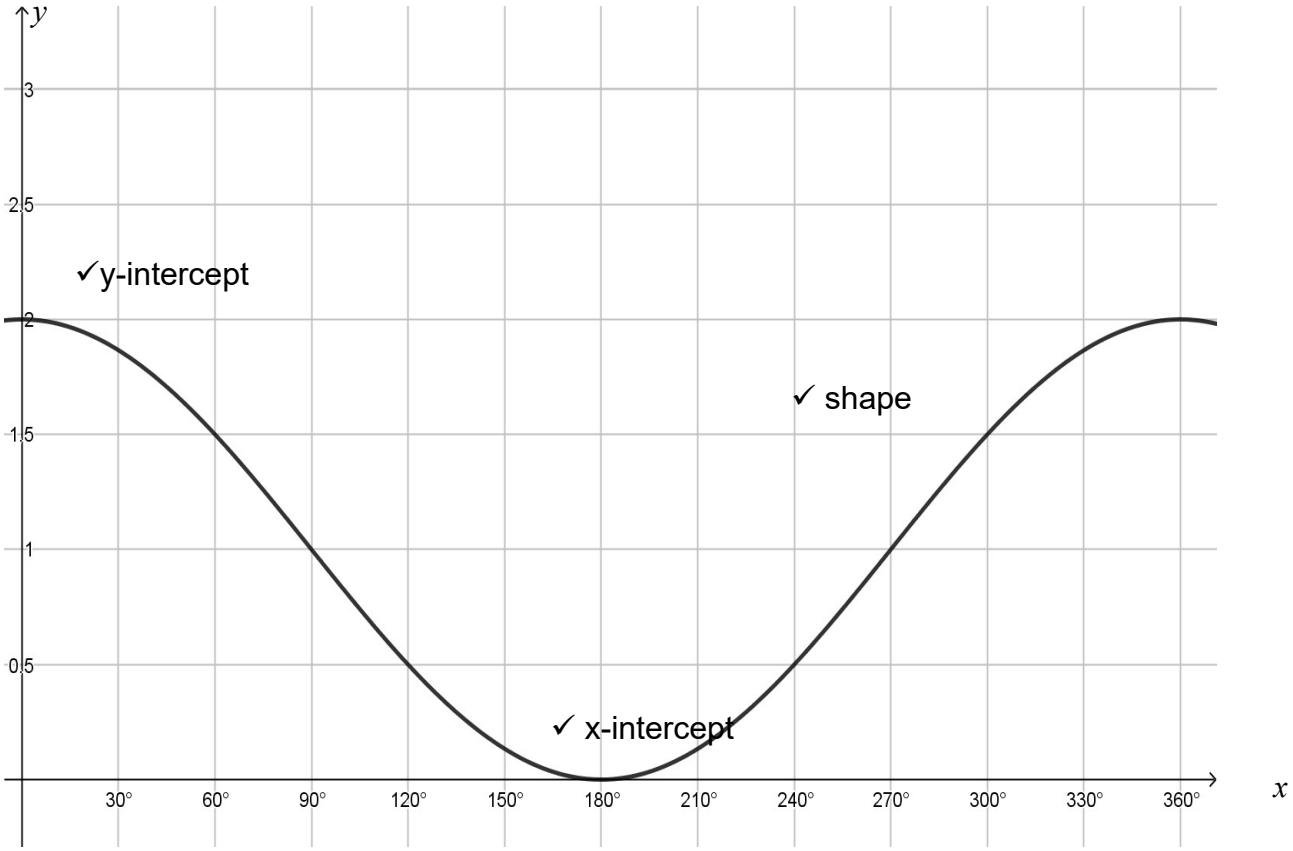
$$\sin 30° = \frac{70}{PR} \checkmark$$

$$PR = \frac{70}{\sin 30°} \checkmark$$

$$PR = 140\text{cm} \checkmark$$

(3)

3.6



(3)
[30]

TOTAL: 100

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